OPMIG-07: Metric & Event Extraction

Series: OPMIG | **Notebook:** 7 of 9 | **Created:** December 2025 | **Last Updated:** 01/28/2026

OpenPipeline Migration Series | Notebook 7 of 9
Level: Intermediate to Advanced
Estimated Time: 70 minutes

Table of Contents

Learning Objectives

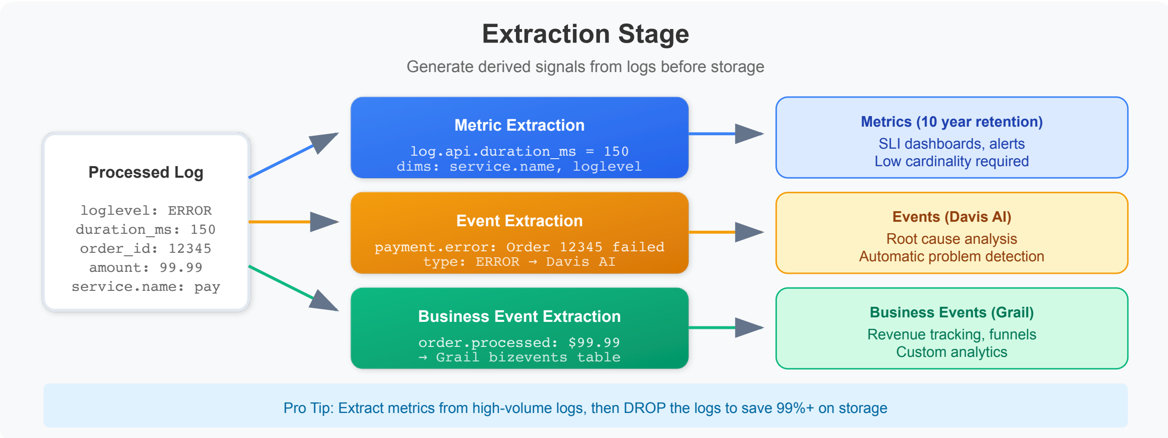
By completing this notebook, you will:

1. Configure metric and event extraction from logs
2. Generate business events for analytics

3. ⭐ **NEW:** Implement SLI metrics using RED methodology (Rate, Errors, Duration)
4. ⭐ **NEW:** Extract business KPIs (revenue, conversion, user journey)
5. ⭐ **NEW:** Apply metric design best practices (cardinality, naming, aggregation)
6. Validate extractions with DQL queries
7. Optimize extraction for cost efficiency

Extraction Stage Overview

The Extraction stage runs **after** Processing and **before** Storage. It generates:



Key Benefits

Benefit	Description
Derived signals	Create metrics from logs without app changes
Cost optimization	Extract metrics, then drop verbose logs
Davis AI integration	Generated events feed into AI analysis
Business visibility	Business events enable analytics dashboards

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Metric Extraction

OpenPipeline supports two types of metrics:

Value Metrics (Gauge)

Extract numeric values as gauge metrics.

Field	Description
Key	Metric name (e.g., `log.payment.amount`)
Value	Numeric field to extract
Dimensions	Fields to use as metric dimensions
Matching Condition	When to apply extraction

Example: Extract payment amount

```
Metric Type: Value
Key: log.payment.amount
Value: amount
Dimensions: service.name, environment, status
Matching: contains(content, "payment") AND isNotNull(amount)
```

Counter Metrics

Count occurrences of matching records.

Field	Description
Key	Metric name (e.g., `log.error.count`)
Dimensions	Fields to use as metric dimensions
Matching Condition	When to count

Example: Count errors

```
Metric Type: Counter
Key: log.errors.count
Dimensions: service.name, error_code
Matching: loglevel == "ERROR"
```

Metric Naming Conventions

```
Format: &lt;source&gt;.&lt;domain&gt;.&lt;name&gt;

Examples:
  log.api.request_count
  log.payment.amount
  log.auth.failed_attempts
  span.checkout.duration_ms
```

Dimension Best Practices

Do	Don't
Low-cardinality fields	High-cardinality IDs
`service.name`	`request_id`
`environment`	`user_id`
`error_code`	`timestamp`

SLI Metric Patterns: RED Methodology ⭐ NEW

The **RED Method** (Rate, Errors, Duration) provides comprehensive service observability.

RED Overview

Metric	Measures	Why
Rate	Requests/sec	Traffic volume, capacity
Errors	Error rate (%)	Reliability, SLA
Duration	Response time (p50, p95, p99)	User experience

Rate: Request Throughput

```
Type: Counter
Key: log.api.request_rate
Dimensions: method, path, status_category, service.name
Matching: isNotNull(path)
```

Errors: Error Rate SLI

Total Requests:

```
Counter: log.api.request_total
Dimensions: service.name, endpoint
```

Error Requests:

```
Counter: log.api.request_errors
Dimensions: service.name, endpoint, error_type
Matching: status &gt;= 500 OR loglevel == "ERROR"
```

Duration: Latency Percentiles

```
Type: Value
Key: log.api.response_time_ms
Value: duration_ms
Dimensions: service.name, endpoint, status_category
```

Latency Buckets (Apdex-style SLI)

Use `fieldsAdd` to create latency buckets:

```
fieldsAdd latency_bucket = if(duration_ms &lt; 100, "fast",
                           else: if(duration_ms &lt; 500, "acceptable",
                           else: if(duration_ms &lt; 1000, "slow",
                           else: "very_slow")))
```

Then create a counter metric:

```
Counter: log.api.latency_bucket_count
Dimensions: latency_bucket, service.name, endpoint
```

Event Extraction

Extract events that Davis AI can analyze for root cause analysis.

Event Configuration

Field	Description
Event Name	Unique event type identifier
Event Description	Template for event description
Matching Condition	When to generate event
Event Type	Category: INFO, AVAILABILITY, ERROR, RESOURCE, CUSTOM

Event Types

Type	Use Case
`INFO`	Informational events
`AVAILABILITY`	Service up/down events
`ERROR`	Error conditions
`RESOURCE`	Resource contention events
`CUSTOM`	Custom event types

Event Description Templates

Use field placeholders in descriptions:

```
Payment failed for order {order_id}: {error_message}
Service {service.name} error rate exceeded threshold
Authentication failure for user {user_id} from IP {client_ip}
```

Example Event Extraction

```
Event Name: payment.failure
Event Description: Payment failed for order {order_id}: {error_message}
Event Type: ERROR
Matching: contains(content, "payment") AND contains(content, "failed")
```

Business Event Extraction

Business events are stored in Grail for business analytics.

Business Event Configuration

Field	Description
Event Type	Business event type name
Event Provider	Source system identifier
Data Fields	Fields to include in event
Matching Condition	When to extract

Example: Order Events

```
Event Type: com.example.order.placed
Event Provider: checkout-service
Data Fields:
  - order_id
  - customer_id
  - total_amount
  - currency
  - timestamp
Matching: contains(content, "order placed successfully")
```

Querying Business Events

Business KPI Extraction ⭐ NEW

Revenue Tracking

Order Revenue:

```
Type: Value
Key: log.revenue.order_amount
Value: amount
Dimensions: currency, service.name, payment_method
```

Conversion Funnels

E-Commerce Funnel:
1. Product View → 2. Add to Cart → 3. Checkout → 4. Complete

Use `fieldsAdd` to categorize funnel stages:

```
fieldsAdd funnel_stage = if(action == "product_view", "1_view",
                         else: if(action == "add_to_cart", "2_cart",
                         else: if(action == "checkout_started", "3_checkout",
                         else: "4_complete")))
```

Feature Usage

```
Counter: log.feature.usage_count
Dimensions: feature_name, feature_enabled, user_segment
```

```
// Query business events extracted from logs
fetch bizevents, from: now() - 24h
| filter event.type == "com.example.order.placed"
| fields timestamp, order_id, customer_id, total_amount
| limit 50
```

```
// Business event analytics - order totals by hour
fetch bizevents, from: now() - 24h
| filter event.type == "com.example.order.placed"
| makeTimeseries {
    order_count = count(),
    total_revenue = sum(total_amount)
  }, interval: 1h
```

Extraction Matching Conditions

Matching conditions determine which records trigger extraction.

Common Patterns

Pattern	Condition
Error logs	`loglevel == "ERROR"`
Specific service	`service.name == "payment-service"`
Contains text	`contains(content, "failed")`
Has field	`isNotNull(order_id)`
Numeric threshold	`duration_ms > 1000`

Combining Conditions

```
// AND - both must match
loglevel == "ERROR" AND service.name == "payment"

// OR - either matches
status == "failed" OR status == "error"

// Complex conditions
(loglevel == "ERROR" OR loglevel == "WARN")
  AND contains(content, "payment")
  AND isNotNull(amount)
```

Field Reference in Conditions

Use any field available after processing:
- Original fields from ingestion
- Parsed fields from processors
- Computed fields from `fieldsAdd`

Practical Examples

Example 1: API Request Metrics

Source Log:

```
2024-12-12T10:30:45 INFO - GET /api/users completed in 150ms, status=200
```

Pipeline Processing:

```
// First, parse the log
parse content, "LD:method ' ' LD:path ' completed in ' INT:duration_ms 'ms, status=' INT:status_code"
```

Metric Extraction 1: Request duration

```
Type: Value
Key: log.api.request_duration_ms
Value: duration_ms
Dimensions: method, path, status_code
Matching: isNotNull(duration_ms)
```

Metric Extraction 2: Request count

```
Type: Counter
Key: log.api.request_count
Dimensions: method, path, status_code
Matching: isNotNull(path)
```

Example 2: Error Event Generation

Source Log:

```
2024-12-12T10:30:45 ERROR PaymentService - Transaction failed: insufficient
funds, orderId=12345
```

Event Extraction:

```
Event Name: payment.transaction.failed
Event Description: Transaction failed for order {order_id}: {error_reason}
Event Type: ERROR
Matching: loglevel == "ERROR" AND contains(content, "Transaction failed")
```

Example 3: Login Business Events

Source Log:

```
2024-12-12T10:30:45 INFO AuthService - User login successful: userId=john123,
ip=192.168.1.100
```

Business Event Extraction:

```
Event Type: com.example.auth.login
Event Provider: auth-service
Data Fields: user_id, client_ip, timestamp
Matching: contains(content, "login successful")
```

Example 4: Combined Extraction

From a single payment log line, extract:

1. **Metric**: `log.payment.amount` (value: amount)
2. **Metric**: `log.payment.count` (counter)
3. **Event**: Payment processed (INFO)
4. **Business Event**: Order transaction

Validating Extractions

After configuring extractions, verify they're working.

```
// List log-extracted metrics
// Note: Use the Dynatrace UI (Observe &gt; Metrics) to browse metrics
// Or use timeseries to query a specific metric:
// timeseries avg_value = avg(log.your_metric_name), from: now() - 24h
```

```
// Query a specific extracted metric
// Replace 'log.api.request_duration_ms' with your metric key
timeseries {
    avg_duration = avg(log.api.request_duration_ms),
    max_duration = max(log.api.request_duration_ms)
  }, from: now() - 2h, interval: 5m
```

```
// View extracted counter metric by dimension
// Replace 'log.api.request_count' with your metric key
timeseries {
    requests = sum(log.api.request_count)
  }, from: now() - 2h, interval: 5m
```

```
// View extracted events
fetch events, from: now() - 24h
| filter isNotNull(event.name)
| summarize {event_count = count()}, by: {event.type, event.name}
| sort event_count desc
```

```
// View specific event details
fetch events, from: now() - 24h
| filter event.name == "payment.transaction.failed"
| fields timestamp, event.name, event.description, event.type
| sort timestamp desc
| limit 25
```

```
// View business events by type
fetch bizevents, from: now() - 24h
| summarize {bizevent_count = count()}, by: {event.type, event.provider}
| sort bizevent_count desc
```

```
// Compare source log volume to extracted metrics
// This helps verify extraction is capturing expected data
fetch logs, from: now() - 1h
| filter contains(content, "payment")
| summarize {log_count = count()}
| fieldsAdd data_source = "logs"

// Then compare with:
// timeseries sum(log.payment.count)
```

Metric Design Best Practices ⭐ NEW

Cardinality Management

Cardinality	Examples	Max Values	Use?
Low	environment, region	< 50	✅ Ideal
Medium	endpoint, error_code	50-500	✅ Good
High	session_id, request_id	500-10K	⚠️ Avoid
Very High	user_id, order_id	> 10K	❌ Never

Cardinality Example:

```
✅ GOOD: 5 services × 3 environments × 4 methods = 60 time series
❌ BAD: 5 services × 100K user_ids = 500K time series
```

Reduction Techniques:

```
// Bucket high-cardinality values
fieldsAdd duration_bucket = if(duration_ms &lt; 100, "&lt;100ms",
                            else: if(duration_ms &lt; 500, "100-500ms",
                            else: "&gt;500ms"))

// Use segments instead of IDs
fieldsAdd user_segment = if(user_tier == "premium", "premium", else: "free")
```

Naming Conventions

Format: `<source>.<domain>.<metric>.<unit>`

Examples:

```
✅ log.api.request_count
✅ log.api.response_time_ms
✅ log.payment.order_amount_usd
✅ log.auth.failed_login_count

❌ apiReqCnt (abbreviated, no unit)
❌ response_time (no source, no unit)
❌ count (no context)
```

Cost Optimization

Metrics vs. Logs:

```
Scenario: 1M requests/day

Option 1: Store logs (35 days)
  - 35M log records × 500 bytes = 17.5 GB
  - Cost: ~$140/month

Option 2: Extract metrics + drop logs
  - 10 time series, 10 years retention = ~10 MB
  - Cost: ~$1/month
  - Savings: 99.3% 🎉
```

When to Extract:
- ✅ SLI tracking (RED)
- ✅ Business KPIs
- ✅ High-volume health checks
- ❌ Debugging (keep logs)
- ❌ Compliance/audit (keep logs)

Best Practices

Metric Extraction

Practice	Reason
Use meaningful metric names	Easier discovery and querying
Limit dimensions (≤5-7)	Avoid metric cardinality explosion
Use low-cardinality dimensions	IDs cause metric bloat
Extract only what you need	Reduces storage costs
Test matching conditions	Ensure correct data selection

Event Extraction

Practice	Reason
Meaningful event names	Easier to find in Davis AI
Descriptive templates	Context for troubleshooting
Appropriate event types	Correct Davis AI handling
Avoid over-extraction	Too many events = noise

Business Event Extraction

Practice	Reason
Consistent event types	Enables analytics
Include all needed fields	Avoids re-processing
Use namespaced types	Prevents collisions
Document event schema	Helps consumers

Cost Optimization Pattern

Extract → Drop → Save

1. Extract metrics and events from verbose logs
2. Drop the original verbose logs
3. Keep only derived signals

This pattern reduces storage while preserving observability.

Complete Extraction Pipeline Example

Pipeline: `checkout-service-logs`

Processing Stage (from OPMIG-06):

```
// Parse order details
parse content, "'orderId=' INT:order_id ','"
| parse content, "'amount=' DOUBLE:amount"
| parse content, "'status=' LD:order_status"
| parse content, "'duration=' INT:duration_ms 'ms'"
```

Metric Extraction 1: Order Amount

```
Type: Value
Key: log.checkout.order_amount
Value: amount
Dimensions: order_status, service.name
Matching: isNotNull(amount)
```

Metric Extraction 2: Order Count

```
Type: Counter
Key: log.checkout.order_count
Dimensions: order_status, service.name
Matching: isNotNull(order_id)
```

Metric Extraction 3: Processing Duration

```
Type: Value
Key: log.checkout.processing_duration_ms
Value: duration_ms
Dimensions: order_status, service.name
Matching: isNotNull(duration_ms)
```

Event Extraction: Order Failure

```
Event Name: checkout.order.failed
Event Description: Order {order_id} failed with status {order_status}
Event Type: ERROR
Matching: order_status == "failed" OR order_status == "error"
```

Business Event: Order Completed

```
Event Type: com.example.checkout.order_completed
Event Provider: checkout-service
Data Fields: order_id, amount, order_status, duration_ms
Matching: order_status == "completed" OR order_status == "success"
```

Next Steps

Now that you can extract metrics and events, continue with:

Notebook	Focus Area
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- OpenPipeline Data Extraction
- Metric Extraction
- Business Event Extraction
- Davis AI Events

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